Read questions carefully and answer, showing all steps of your solution

Final Answers with no clear solution get ZERO

Question 1 :

Use central difference approximations to estimate the second derivative $O(h^2)$ of $y=e^{-x}\cos x$, at $x=\pi/4$, using a step size $h=\pi/12$.

Question 2:

Use the **best numerical differentiation approximation** to find the rocket's **velocity** and **acceleration** at t=25 seconds and t=125 seconds.

The data provided is:

t (s)	25	50	75	100	125
y (km)	32	58	78	92	100

Question 3:

Given the following system of linear equations:

$$x + 2y - z = 4$$
$$2x + 4y + 3z = 5$$
$$-x - 2y + kz = -1$$

Find the value of k for which the system of equations has:

1. No solution.
2. An infinite number of solutions.
3. A unique solution.

Question 4:

Given the following data:

x	1	2	3	4	5	6
f(x)	-2	1	10	29	62	113

1. Construct the Newton's forward difference interpolation polynomial.
2. Use the polynomial to find the approximate value of the **first derivative**, f'(1), at the first data point.
3. Use the polynomial to find the approximate value of the **second derivative**, f''(3), at an interior point.
4. Use the polynomial to find the approximate value of the **third derivative**, f'''(4), at another interior point.

Question 5 :

Given the differential equation $\frac{dy}{dx} = 2x - y$, with the initial condition $y(0)=1$.

Using **a single step** of the **RK4 method**, find the approximate solution at $x=0.2$ with a step size of $h=0.2$.

Question 6 :

x	1	2	3	4	5	6	7	8
f(x)	2	5	10	17	26	37	50	65

Use numerical integration to find the approximate value of the definite integral $\int_0^7 f(x)\,dx$ using:

1. **Simpson's 1/3 Rule**
2. **Simpson's 3/8 Rule**

Question 7 :

Determine the **Gauss Quadrature Formula** of the following integral:

$$\int_{-1}^{1} (2x^2 + 5)\,dx$$

Use the provided nodes $x_0 = -1, x_1 = 0, x_2 = 1$.

Question 8 :

IF we fit a 2nd order polynomial for 8 values, the correlation coefficient is 0.9880, $\sum(y_i - \bar{y})^2$ is 2500 calculate :

1. The Standard error of estimate based on regression polynomial
2. Standard deviation.

Question 9 :

Given the function:

$$f(x, y) = 2x^2 + 5y^2 + 4xy - 3x - 2y$$

Use the **Optimal Steepest Ascent Method** to maximize this function, starting from the point (1,-1). Determine the next point.